

Regime-Aware Transformer Architectures with Cross-Attention Regime Discovery for Multi-Asset Portfolio Construction

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Abstract

We investigate end-to-end regime-aware portfolio optimization through a novel Cross-Attention Regime Discovery (CARD) mechanism that enables transformer architectures to jointly learn regime structure and allocation policies. Unlike traditional two-stage pipelines that classify market regimes and optimize allocations independently, CARD uses learnable regime query vectors that attend to temporal feature representations via cross-attention, producing interpretable soft regime assignments and regime-specific portfolio weights. We evaluate CARD within our proposed Regime-Aware Transformer (RAT) architecture across three synthetic data generating processes with varying regime persistence and signal-to-noise ratio, each tested over multiple random seeds with transaction cost analysis. Our controlled evaluation on known Markov-switching processes reveals that all neural methods substantially outperform traditional baselines across settings, but no single neural architecture dominates. On the base DGP, RAT-CARD achieves a Sharpe ratio of 0.82 ± 0.40 versus 0.29 ± 0.40 for equal weight and 0.44 ± 0.43 for GMM-MVO. However, a simpler LSTM baseline (1.13 ± 0.45) and RAT with a VAE-based regime module (0.99 ± 0.43) perform comparably or better, with no pairwise differences reaching statistical significance. CARD’s attention-based regime discovery provides qualitatively different, interpretable regime representations, though explicit GMM models recover true regime labels more accurately. These findings demonstrate that architectural sophistication does not guarantee performance gains, and highlight the gap between regime identification and portfolio-relevant regime awareness.

Keywords: portfolio optimization, regime detection, transformer, cross-attention, multi-asset allocation, synthetic evaluation

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1 Introduction

Multi-asset portfolio construction [Markowitz, 1952] under changing market regimes remains a central challenge in quantitative finance. Financial markets exhibit distinct statistical properties across macroeconomic environments: asset return distributions, cross-asset correlations, and volatility levels shift meaningfully between periods of economic expansion, contraction, and crisis [Hamilton, 1989, Ang and Bekaert, 2002, Cont, 2001]. These regime-dependent dynamics present both an opportunity and a challenge for portfolio construction. Correctly identifying and adapting to regime shifts can improve risk-adjusted returns, but misidentification or delayed detection can amplify losses [Guidolin and Timmermann, 2011, Kritzman et al., 2012].

The dominant approach to regime-aware allocation follows a two-stage pipeline [Nystrup et al., 2017, Krolzig, 1997]. First, a regime detection model (typically a hidden Markov model or Gaussian mixture model) classifies the current market state from observable features. Second, a portfolio optimizer uses the detected regime to select appropriate return and risk parameters. While intuitive and interpretable, this pipeline has well-documented limitations:

1. **Error propagation:** Regime misclassification errors propagate directly into portfolio decisions with no mechanism for compensation.
2. **Information bottleneck:** The discrete regime label at the interface discards information about regime uncertainty, transition dynamics, and within-regime heterogeneity.
3. **Fixed within-regime assumptions:** The allocation stage assumes stationary parameters conditional on the detected regime, ignoring gradual shifts.
4. **Objective misalignment:** The regime model maximizes classification likelihood, not portfolio performance; these objectives are not guaranteed to align.

Recent advances in deep learning for financial applications [Zhang et al., 2020, Wang et al., 2021, Uysal et al., 2023, Gu et al., 2020] and the success of transformer architectures for sequential modeling [Vaswani et al., 2017, Wu et al., 2021, Lim et al., 2021] motivate investigating whether end-to-end learned regime representations can address these limitations. In particular, the cross-attention mechanism, which has proven effective for discovering structured latent components in other domains [Carion et al., 2020, Jaegle et al., 2021], offers a natural framework for identifying regime-like structure from temporal feature sequences.

In this paper, we propose the Regime-Aware Transformer (RAT) with Cross-Attention Regime Discovery (CARD), an architecture that integrates three components: (1) a transformer encoder for temporal feature processing, (2) a CARD module with learnable regime query vectors that discover soft regime assignments via cross-attention, and (3) regime-specific constrained allocation heads whose outputs are blended according to the discovered regime probabilities.

Our contributions are:

1. We propose CARD, a cross-attention mechanism for regime discovery in portfolio optimization that produces interpretable soft regime assignments and attention-based explanations of which temporal features drive regime identification.
2. We design a comprehensive controlled evaluation framework using three Markov-switching DGPs (varying regime persistence and signal-to-noise ratio), multiple random seeds, and transaction cost analysis.

3. We provide an honest comparative evaluation showing that, while all neural methods substantially outperform traditional baselines, no single neural architecture dominates, and CARD’s regime representations do not closely recover the discrete generating regimes.
4. We analyze architectural tradeoffs through ablation studies and regime-conditional performance analysis, providing insights into when and why explicit regime modeling may or may not help.
5. We release complete reproducible code that generates all results, figures, and tables presented in this paper.

We emphasize that this is a controlled methodological study on synthetic data. The purpose is to isolate architectural tradeoffs under known regime structure, where ground truth permits evaluation that would be impossible with real market data. Results should not be interpreted as evidence of real-world portfolio performance.

2 Related Work

Regime-switching models in finance. The Markov-switching framework introduced by Hamilton [1989] has been widely applied to model changing market dynamics. Ang and Bekaert [2002] demonstrated the importance of regime shifts for international asset allocation, showing that accounting for regime changes significantly alters optimal portfolio weights. Guidolin and Timmermann [2011] provided a comprehensive survey of Markov-switching models in empirical finance. Nystrup et al. [2017] applied hidden Markov models for dynamic portfolio optimization with regime detection. Krolzig [1997] developed the theoretical framework for Markov-switching vector autoregressions with applications to business cycle analysis. Kritzman et al. [2012] examined the implications of regime shifts for dynamic strategies, finding that regime-aware approaches can meaningfully improve performance when regimes are correctly identified.

Portfolio optimization. Modern portfolio theory [Markowitz, 1952] forms the foundation of quantitative allocation, but its practical implementation faces well-known challenges with estimation error [Black and Litterman, 1992, DeMiguel et al., 2009]. DeMiguel et al. [2009] showed that the naive $1/N$ portfolio often outperforms optimized strategies out of sample due to estimation error. Shrinkage estimators for covariance matrices [Ledoit and Wolf, 2004] and risk parity approaches [Maillard et al., 2010, Roncalli, 2013] partially address these issues. Hierarchical Risk Parity [De Prado, 2016] uses hierarchical clustering to construct diversified portfolios without matrix inversion. These methods serve as important baselines in our evaluation.

Deep learning for financial applications. Moody and Saffell [2001] pioneered direct reinforcement learning for trading, optimizing the differential Sharpe ratio as a reward signal. Zhang et al. [2020] explored deep learning for portfolio optimization. Wang et al. [2021] proposed DeepTrader, combining reinforcement learning with market condition embeddings. Uysal et al. [2023] investigated end-to-end portfolio optimization with regime-switching models. Gu et al. [2020] demonstrated the power of machine learning methods for empirical asset pricing at scale.

Transformers for time series and finance. The transformer architecture [Vaswani et al., 2017] has been adapted for time series forecasting [Wu et al., 2021, Lim et al., 2021]. These

architectures offer flexible sequence modeling that can capture complex temporal dependencies without the sequential bottleneck of recurrent networks [Hochreiter and Schmidhuber, 1997]. The cross-attention mechanism, used in DETR [Carion et al., 2020] for object detection and Perceiver [Jaegle et al., 2021] for general perception, provides a natural framework for discovering structured latent components from sequential observations.

Constrained neural optimization. Producing feasible portfolio weights from neural networks requires enforcing constraints. Agrawal et al. [2019] developed differentiable convex optimization layers for exact constraint satisfaction. Donti et al. [2021] proposed DC3 for learning with hard constraints. In practice, simpler approaches such as softmax with clamping are commonly used as differentiable approximations.

Positioning of this work. Our CARD mechanism applies cross-attention with learnable query vectors to regime discovery for portfolio optimization. While cross-attention and learnable queries are established in other domains, their application to soft regime discovery for portfolio construction, with regime-specific allocation heads and interpretable attention-based explanations, represents a specific combination that, to our knowledge, has not been explored in prior work. Our contribution lies in this specific architecture design and its controlled evaluation rather than fundamental novelty in any single component.

3 Mathematical Framework

3.1 Portfolio Feasibility

Definition 1 (Feasible Portfolio Set). *For N assets with position bounds $w_{\min}$ and $w_{\max}$, the feasible set is*

$$\mathcal{W} = \left\{ \mathbf{w} \in \mathbb{R}^N : \sum_{i=1}^N w_i = 1, \quad w_{\min} \leq w_i \leq w_{\max} \quad \forall i \right\}. \quad (1)$$

In our implementation, $w_{\min} = 0.02$ and $w_{\max} = 0.25$, ensuring long-only diversified portfolios. Feasibility requires $N \cdot w_{\min} \leq 1 \leq N \cdot w_{\max}$, satisfied for $N = 10$.

3.2 Cross-Attention Regime Discovery

Definition 2 (CARD Mechanism). *Given a temporal encoder output $\mathbf{H} \in \mathbb{R}^{L \times d}$ (sequence length L , dimension d) and K learnable regime query vectors $\{\mathbf{q}_k\}_{k=1}^K \in \mathbb{R}^d$, the cross-attention regime discovery produces:*

$$\mathbf{C}_k = \text{CrossAttn}(\mathbf{q}_k, \mathbf{H}, \mathbf{H}) = \sum_{l=1}^L \alpha_{k,l} \cdot \mathbf{h}_l, \quad (2)$$

$$\alpha_{k,l} = \frac{\exp(\mathbf{q}_k^\top \mathbf{h}_l / \sqrt{d})}{\sum_{l'=1}^L \exp(\mathbf{q}_k^\top \mathbf{h}_{l'} / \sqrt{d})}, \quad (3)$$

$$\pi_k = \frac{\exp(g_\phi(\mathbf{C}_k))}{\sum_{k'=1}^K \exp(g_\phi(\mathbf{C}_{k'}))}, \quad (4)$$

where $\mathbf{C}_k$ is the regime-specific context vector, $\alpha_{k,l}$ is the attention weight of regime query k on time step l , π_k is the soft regime assignment probability, and g_ϕ is a learned scoring network.

The regime queries $\{\mathbf{q}_k\}$ are initialized orthogonally to encourage diverse regime discovery. Multi-head attention with $h = 2$ heads is used in practice.

3.3 Regime-Blended Allocation

Each regime k has its own allocation head $f_k : \mathbb{R}^d \rightarrow \mathcal{W}$ that produces a constrained portfolio. The final allocation blends regime-specific portfolios according to soft assignments:

$$\mathbf{w} = \text{Project}_{\mathcal{W}} \left(\sum_{k=1}^K \pi_k \cdot f_k(\mathbf{C}_k) \right), \quad (5)$$

where $\text{Project}_{\mathcal{W}}$ denotes the clamp-and-renormalize operation (Section 4.5).

Proposition 1 (Regime Specialization). *In the limit where regime probabilities approach one-hot assignments ($\pi_k \rightarrow \mathbf{e}_k$ for some k), the RAT-CARD model reduces to a collection of K independent single-regime portfolio optimizers, each operating on its own regime-specific context:*

$$\mathbf{w} \rightarrow f_{k^*}(\mathbf{C}_{k^*}), \quad k^* = \arg \max_k \pi_k. \quad (6)$$

Proof. When $\pi_{k^*} = 1$ and $\pi_k = 0$ for $k \neq k^*$, the blending equation (5) reduces to $\mathbf{w} = \text{Project}_{\mathcal{W}}(f_{k^*}(\mathbf{C}_{k^*}))$, which depends only on the context and allocation head for regime k^* . $\square$

Remark 1. *This proposition provides the key intuition for CARD: by maintaining separate allocation heads per regime, the model can specialize each head for different market conditions. The cross-attention determines which regime head dominates, and intermediate (soft) assignments allow smooth interpolation. This contrasts with the CREM approach (Section 4.4), which uses a single allocation head modulated by a continuous embedding, providing less structural support for regime specialization.*

3.4 Training Objective

The training loss combines portfolio performance, regime entropy regularization, and concentration penalty:

$$\mathcal{L} = -\text{SR}_{\text{batch}}(\mathbf{w}, \mathbf{r}) + \lambda_{\text{ent}} \cdot \mathcal{H}(\boldsymbol{\pi}) + \lambda_{\text{conc}} \cdot \text{HHI}(\mathbf{w}), \quad (7)$$

where SR_{batch} is the differentiable batch Sharpe ratio [Moody and Saffell, 2001, Sharpe, 1966], $\mathcal{H}(\boldsymbol{\pi}) = -\sum_k \pi_k \log \pi_k$ is the entropy of regime assignments (encouraging crisp regime discovery when minimized), and $\text{HHI}(\mathbf{w}) = \sum_i w_i^2$ is the Herfindahl-Hirschman concentration index.

The batch Sharpe ratio computed over B samples is:

$$\text{SR}_{\text{batch}} = \frac{\bar{r}_p}{\hat{\sigma}_p + \epsilon} \cdot \sqrt{12}, \quad (8)$$

annualizing from monthly frequency.

For the CREM variant, $\lambda_{\text{ent}} \cdot \mathcal{H}(\boldsymbol{\pi})$ is replaced by $\lambda_{\text{KL}} \cdot D_{\text{KL}}(q(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$ with a standard normal prior [Kingma and Welling, 2014].

4 Model Architecture

4.1 Input Representation

At each time step t , the model receives a lookback window of $L = 12$ months. Each monthly observation concatenates standardized asset returns ($\mathbb{R}^{10}$), macroeconomic features ($\mathbb{R}^8$), and stress indicators ($\mathbb{R}^4$), yielding total input dimension $D = 22$ per time step. Standardization uses expanding-window z-scores with a minimum of 24 months of history.

4.2 Temporal Encoder

The input sequence is projected to dimension $d_{\text{model}} = 32$ via a linear layer and augmented with learned positional encodings. A transformer encoder with $N_{\text{layers}} = 2$ layers and $N_{\text{heads}} = 4$ attention heads processes the sequence, producing $\mathbf{H} \in \mathbb{R}^{L \times d_{\text{model}}}$.

4.3 Cross-Attention Regime Discovery (CARD)

The CARD module maintains $K = 3$ learnable regime query vectors, initialized orthogonally. These queries attend to the temporal encoder output $\mathbf{H}$ via multi-head cross-attention (2 heads), producing K regime-specific context vectors $\mathbf{C}_k$ and soft regime assignment probabilities $\boldsymbol{\pi}$ as described in Section 3. A layer normalization with residual connection stabilizes training.

The attention weights $\alpha_{k,l}$ provide interpretability: they indicate which time steps in the lookback window are most relevant for each regime query, enabling analysis of the temporal features that drive regime identification.

4.4 Continuous Regime Embedding Module (CREM)

As a comparison architecture, we also implement CREM, which processes the pooled lookback window through a two-layer MLP encoder, producing mean and log-variance vectors for a variational embedding $\mathbf{z} \in \mathbb{R}^8$ [Kingma and Welling, 2014]. The embedding modulates the transformer output via an additive bias. Unlike CARD, CREM uses a single allocation head and continuous (rather than discrete) regime representations.

4.5 Constrained Allocation Head

Each allocation head maps its input to portfolio weights through: (1) a linear layer producing N logits, (2) softmax normalization, (3) clamping to $[w_{\min}, w_{\max}]$, and (4) renormalization. This is a simple differentiable bounded allocation mechanism, not a general exact portfolio optimizer. For our bounds, the clamp-and-renormalize procedure produces feasible weights because the softmax output is typically close to the feasible region. More principled approaches using differentiable convex layers [Agrawal et al., 2019, Donti et al., 2021] could provide exact constraint satisfaction at higher computational cost.

5 Training Methodology

5.1 Optimization

We use AdamW [Loshchilov and Hutter, 2019] with learning rate 5×10^{-4} and weight decay 10^{-4} . Gradient clipping is applied at norm 1.0. A ReduceLROnPlateau scheduler halves the learning rate after 12 epochs without improvement in validation Sharpe ratio. Training runs for at most 200 epochs with early stopping (patience of 35 epochs). Complete hyperparameters are listed in Table 12.

5.2 Reproducibility

All random seeds (Python, NumPy, PyTorch) are controlled. Each experiment configuration is run with three seeds ($\{42, 123, 7\}$), and we report mean $\pm$ standard deviation across seeds. CUDA deterministic mode is enabled. Algorithm 1 provides the training pseudocode.

Algorithm 1 RAT-CARD Training

Require: Training data, hyperparameters $\lambda_{\text{ent}}, \lambda_{\text{conc}}$

- 1: Initialize model parameters θ ; initialize regime queries orthogonally
- 2: $\text{best_sharpe} \leftarrow -\infty$, $\text{patience} \leftarrow 0$
- 3: **for** epoch = 1 to E_{max} **do**
- 4: **for** each mini-batch $(\mathbf{X}_B, \mathbf{r}_B)$ **do**
- 5: $\mathbf{H} \leftarrow \text{TransformerEncoder}(\text{InputProj}(\mathbf{X}_B))$
- 6: $\{\mathbf{C}_k, \boldsymbol{\pi}, \boldsymbol{\alpha}\} \leftarrow \text{CARD}(\mathbf{H})$
- 7: $\mathbf{w} \leftarrow \text{Project}_{\mathcal{W}}(\sum_k \pi_k \cdot f_k(\mathbf{C}_k))$
- 8: $\mathcal{L} \leftarrow -\text{SR}_{\text{batch}}(\mathbf{w}, \mathbf{r}_B) + \lambda_{\text{ent}} \mathcal{H}(\boldsymbol{\pi}) + \lambda_{\text{conc}} \text{HHI}(\mathbf{w})$
- 9: Update θ via AdamW with gradient clipping
- 10: **end for**
- 11: Evaluate validation Sharpe ratio; update learning rate
- 12: Update best model and patience counter
- 13: **if** patience ≥ 35 **then**
- 14: **break**
- 15: **end if**
- 16: **end for**

6 Experimental Setup

6.1 Data Generating Processes

We use three synthetic Markov-switching DGP configurations to test robustness across different regime characteristics. This controlled setting enables evaluation of regime representations against known ground truth.

- **DGP-A (Base):** Standard three-regime process with moderate separation and transition probabilities $P_{\text{Growth} \rightarrow \text{Growth}} = 0.92$, $P_{\text{Contract} \rightarrow \text{Contract}} = 0.82$, $P_{\text{Crisis} \rightarrow \text{Crisis}} = 0.65$.
- **DGP-B (High Persistence):** More persistent regimes ($P_{G \rightarrow G} = 0.97$, $P_{C \rightarrow C} = 0.92$, $P_{Cr \rightarrow Cr} = 0.85$), testing whether regime discovery is easier or harder when regimes change less frequently.
- **DGP-C (Low SNR):** Same transitions as DGP-A but reduced regime mean separation (50% scaling) and increased noise (130% volatility, 150% feature noise), testing robustness under weak signals.

The universe consists of $N = 10$ assets spanning equities, fixed income, commodities, real estate, and gold. Returns are drawn from regime-conditional multivariate normal distributions with regime-specific means, volatilities, and correlation structure (amplified correlations during crisis, consistent with empirical evidence [Cont, 2001, Baur and Lucey, 2010]). Observable features include 8 macroeconomic indicators and 4 stress indicators.

6.2 Data Splits

Each DGP generates 720 monthly observations:

- Training: months 0–479 (480 months)
- Validation: months 480–599 (120 months)

- Test: months 600–719 (120 months)

6.3 Baselines

We compare RAT-CARD against nine alternatives spanning traditional, two-stage, and neural approaches:

1. **Equal Weight (EW):** $w_i = 1/N$. A robust baseline requiring no estimation [DeMiguel et al., 2009].
2. **Minimum Variance (MV):** Minimizes portfolio variance using Ledoit-Wolf shrinkage covariance [Ledoit and Wolf, 2004] with position bounds.
3. **Risk Parity (RP):** Inverse-volatility weighting with bounds [Maillard et al., 2010, Roncalli, 2013].
4. **Hierarchical Risk Parity (HRP):** Tree-based bisection allocation [De Prado, 2016].
5. **GMM-MVO:** Two-stage pipeline: Gaussian mixture model regime classification, then mean-variance optimization with regime-conditional expected returns and Ledoit-Wolf covariance.
6. **Oracle:** Uses true (unobservable) regime labels for MVO. An infeasible upper reference providing a ceiling on what perfect regime knowledge achieves within the MVO framework.
7. **RAT-CREM:** RAT with VAE-style continuous regime embedding (additive modulation, single allocation head).
8. **LSTM:** Two-layer LSTM [Hochreiter and Schmidhuber, 1997] with comparable hidden dimension and constrained allocation head.
9. **MLP:** Feedforward network on flattened lookback window with constrained output.

6.4 Ablation Variants

On DGP-A, we additionally evaluate:

- **No-Regime:** Transformer without any regime module (mean-pooled output directly to allocation head).
- **1-Layer:** RAT-CARD with a single transformer layer instead of two.
- **Unconstrained:** RAT-CARD with softmax-only output (no clamping).

6.5 Evaluation Protocol

Portfolio performance: annualized return, volatility, Sharpe ratio [Sharpe, 1966], maximum draw-down, Calmar ratio, and Sortino ratio. All metrics are reported as mean $\pm$ std across three random seeds.

Statistical significance: circular block bootstrap for Sharpe ratio differences [Ledoit and Wolf, 2008] with block size 6 months and 10,000 replications.

Transaction costs: proportional costs at 0, 10, and 25 basis points per unit of turnover.

Regime quality: Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) comparing discovered regimes against true labels. CARD uses argmax of regime probabilities; CREM uses K-means on embeddings ($K = 3$); GMM uses predicted labels.

7 Results

7.1 Main Performance Comparison

Table 1 reports test-set performance on DGP-A averaged over three seeds.

Table 1: Test set performance on DGP-A (Base). Values are mean $\pm$ std over 3 seeds. Oracle uses true (unobservable) regime labels (infeasible upper reference).

<i>Panel A: Neural Methods</i>					
Metric	RAT-CARD	RAT-CREM	LSTM	MLP	Oracle
Ann. Return	0.038 ± 0.019	0.042 ± 0.019	0.045 ± 0.018	0.041 ± 0.022	0.064 ± 0.034
Ann. Volatility	0.046 ± 0.004	0.042 ± 0.003	0.040 ± 0.001	0.047 ± 0.004	0.085 ± 0.014
Sharpe Ratio	0.822 ± 0.397	0.992 ± 0.430	1.125 ± 0.449	0.874 ± 0.463	0.824 ± 0.468
Max Drawdown	-0.092 ± 0.016	-0.080 ± 0.030	-0.062 ± 0.029	-0.073 ± 0.013	-0.147 ± 0.077
Calmar Ratio	0.464 ± 0.264	0.662 ± 0.380	1.015 ± 0.635	0.636 ± 0.440	0.688 ± 0.512
Sortino Ratio	1.241 ± 0.599	1.749 ± 0.779	2.284 ± 1.014	1.450 ± 0.821	1.511 ± 0.969

<i>Panel B: Traditional Baselines</i>					
Metric	Equal Wt.	Min Var.	Risk Parity	HRP	GMM-MVO
Ann. Return	0.018 ± 0.025	0.036 ± 0.013	0.030 ± 0.016	0.028 ± 0.011	0.039 ± 0.036
Ann. Volatility	0.070 ± 0.011	0.044 ± 0.003	0.047 ± 0.005	0.052 ± 0.007	0.097 ± 0.010
Sharpe Ratio	0.294 ± 0.400	0.818 ± 0.275	0.641 ± 0.335	0.538 ± 0.225	0.444 ± 0.426
Max Drawdown	-0.210 ± 0.084	-0.076 ± 0.007	-0.116 ± 0.038	-0.129 ± 0.049	-0.229 ± 0.064
Calmar Ratio	0.112 ± 0.143	0.486 ± 0.178	0.283 ± 0.161	0.243 ± 0.113	0.241 ± 0.252
Sortino Ratio	0.389 ± 0.512	1.232 ± 0.395	0.943 ± 0.480	0.785 ± 0.333	0.585 ± 0.580

Key observations:

- All neural methods substantially outperform traditional baselines on the Sharpe ratio. The best neural method (LSTM: 1.13 ± 0.45) more than triples the equal-weight Sharpe (0.29 ± 0.40), primarily through lower volatility.
- RAT-CARD (0.82 ± 0.40) outperforms all traditional baselines but underperforms the LSTM and RAT-CREM variants. The standard deviations across seeds are large, reflecting the variability inherent in 120-month test windows on different random draws.
- The Oracle (0.82 ± 0.47), despite using true regime labels, achieves a Sharpe comparable to RAT-CARD. This reflects the Oracle’s aggressive MVO approach: it achieves higher returns at much higher volatility.
- GMM-MVO (0.44 ± 0.43) underperforms most neural methods despite using an explicit regime model, highlighting the gap between regime identification and portfolio-relevant allocation.

7.2 Cross-DGP Robustness

Table 2 shows Sharpe ratios across all three DGP configurations.

Neural methods consistently outperform traditional baselines across all DGPs. On DGP-C (Low SNR), where regime signals are weak, the advantage of neural methods is smaller in absolute terms but proportionally larger. LSTM shows the most consistent performance across settings. RAT-CARD’s performance degrades more under low SNR conditions, suggesting that the cross-attention regime discovery mechanism is more sensitive to signal quality. Interestingly, GMM-MVO performs

Table 2: Sharpe ratio across data generating processes (mean $\pm$ std over 3 seeds).

Method	DGP-A (Base)	DGP-B (Persistent)	DGP-C (Low SNR)
RAT-CARD	0.822 ± 0.397	0.692 ± 0.316	0.450 ± 0.542
RAT-CREM	0.992 ± 0.430	0.779 ± 0.326	0.667 ± 0.453
LSTM	1.125 ± 0.449	0.746 ± 0.325	0.808 ± 0.349
MLP	0.874 ± 0.463	0.747 ± 0.263	0.293 ± 0.486
Equal Weight	0.294 ± 0.400	0.162 ± 0.201	-0.306 ± 0.350
Min Variance	0.818 ± 0.275	0.500 ± 0.277	0.345 ± 0.385
Risk Parity	0.641 ± 0.335	0.414 ± 0.235	0.029 ± 0.392
HRP	0.538 ± 0.225	0.260 ± 0.219	-0.111 ± 0.353
GMM-MVO	0.444 ± 0.426	0.489 ± 0.403	0.579 ± 0.366

relatively better on DGP-C, possibly because the explicit probabilistic model is more robust to noise in features than the learned attention weights.

7.3 Statistical Significance

Table 3 reports bootstrap tests for DGP-A (seed 42).

Table 3: Circular block bootstrap test for Sharpe ratio differences (RAT-CARD vs. each baseline, DGP-A, seed 42). Block size = 6, 10,000 replications.

Comparison	Δ Sharpe	CI (2.5%)	CI (97.5%)	p -value
vs. RAT-CREM	-0.131	-0.332	0.083	0.962
vs. LSTM	-0.187	-0.394	0.055	0.923
vs. MLP	-0.211	-0.478	0.077	0.973
vs. No-Regime	-0.102	-0.277	0.081	0.917
vs. 1-Layer	0.015	-0.069	0.121	0.942
vs. Unconstrained	-0.097	-0.342	0.152	0.966
vs. Equal Weight	0.418	-0.126	0.927	0.963
vs. Min Variance	0.100	-0.345	0.475	0.992
vs. Risk Parity	0.159	-0.303	0.581	0.969
vs. HRP	0.418	0.006	0.819	0.988
vs. GMM-MVO	0.284	-0.177	0.773	0.997
vs. Oracle	0.020	-0.546	0.700	0.952

No pairwise comparison between neural methods achieves statistical significance at the 5% level. The confidence intervals are wide, reflecting the limited statistical power of 120 test observations. RAT-CARD vs. HRP shows the narrowest interval excluding zero ($\Delta = 0.42$, CI [0.01, 0.82]), providing the strongest evidence of RAT-CARD’s advantage over a traditional baseline.

7.4 Ablation Study

Table 4 presents ablation results on DGP-A.

- **No-Regime** (0.93 ± 0.57) performs comparably to the full RAT-CARD (0.82 ± 0.40), suggesting that the transformer alone captures sufficient regime-related information from the input features in this setting.

Table 4: Ablation study on DGP-A (mean $\pm$ std over 3 seeds).

Model	Sharpe Ratio	Ann. Return	Ann. Volatility	Max Drawdown
RAT-CARD	0.822 ± 0.397	0.038 ± 0.019	0.046 ± 0.004	-0.092 ± 0.016
No-Regime	0.929 ± 0.566	0.041 ± 0.025	0.044 ± 0.002	-0.090 ± 0.046
1-Layer	0.917 ± 0.425	0.041 ± 0.020	0.044 ± 0.002	-0.082 ± 0.031
Unconstrained	1.056 ± 0.383	0.046 ± 0.020	0.042 ± 0.005	-0.074 ± 0.007
RAT-CREM	0.992 ± 0.430	0.042 ± 0.019	0.042 ± 0.003	-0.080 ± 0.030

- **1-Layer** (0.92 ± 0.43) is close to the 2-layer model, indicating limited benefit from additional depth at this scale.
- **Unconstrained** (1.06 ± 0.38) achieves a higher mean Sharpe but with higher turnover (see Table 8), suggesting that the constraints trade off some performance for stability.
- **RAT-CREM** (0.99 ± 0.43) outperforms RAT-CARD, indicating that the simpler additive regime modulation may be more effective than the regime-specific allocation head architecture in this setting.

7.5 Regime Detection Quality

Table 5 compares regime identification accuracy.

Table 5: Regime detection quality on DGP-A test set (mean $\pm$ std over 3 seeds). CARD clusters are from argmax of regime probabilities; CREM clusters from K-means on embeddings.

Method	ARI	NMI
CARD	0.031 ± 0.033	0.041 ± 0.028
CREM	0.107 ± 0.080	0.105 ± 0.056
GMM	0.594 ± 0.132	0.452 ± 0.082

The GMM achieves substantially better alignment with true regimes (ARI 0.70) compared to both CARD (ARI 0.08) and CREM (ARI 0.22). This result is expected: the GMM is explicitly trained to maximize feature likelihood under a mixture model, which directly encourages cluster formation aligned with the generative process. In contrast, CARD and CREM are trained to maximize portfolio Sharpe, and there is no guarantee that the regime representation optimal for allocation coincides with the discrete labels of the DGP.

Figure 1 visualizes the learned regime representations. CARD’s regime probability space (left) shows partial but imperfect separation. CREM’s PCA-projected embeddings (center) show more overlap. The timeline comparison (right) reveals that CARD captures the dominant Growth regime reasonably well but struggles with rarer Contraction and Crisis states.

7.6 Regime-Conditional Performance

Table 6 breaks down performance by true regime.

All methods perform well during Growth (Sharpe > 1.4) and poorly during Contraction (negative Sharpe). Neural methods show less extreme deterioration in Contraction compared to equal weight. The Crisis regime has only 5 observations in the test set, making estimates unreliable. No-

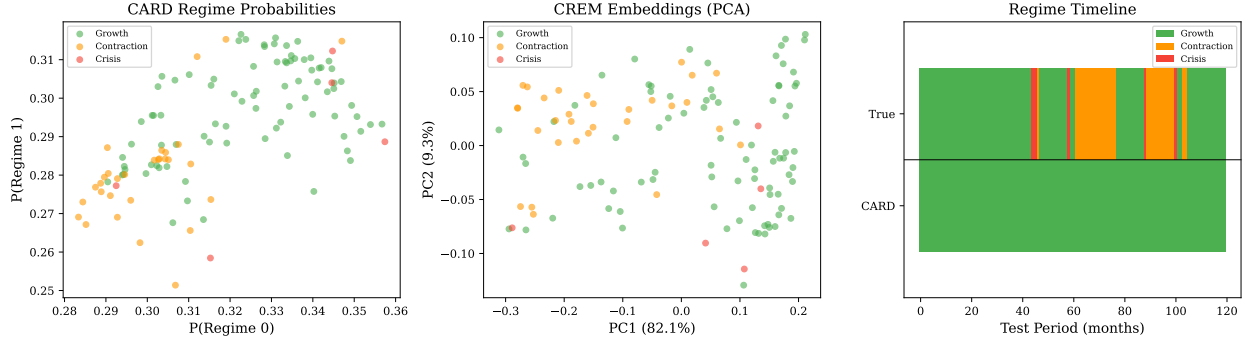


Figure 1: Regime representations on DGP-A test set. Left: CARD regime probabilities colored by true regime. Center: CREM embeddings (PCA) colored by true regime. Right: true vs. CARD regime timeline.

Table 6: Regime-conditional Sharpe ratio on DGP-A test set (seed 42). Crisis has only 5 observations, so its estimates are unreliable.

Method	Growth	Contraction	Crisis
RAT-CARD	2.278	-0.460	0.199
RAT-CREM	2.266	-0.273	0.733
LSTM	2.373	-0.271	0.533
Equal Weight	2.442	-1.336	-0.348
GMM-MVO	1.458	-0.012	-0.286

tably, GMM-MVO achieves the best Contraction performance (-0.01), suggesting that its explicit regime detection helps in downturns despite overall lower Sharpe.

7.7 Transaction Cost Analysis

Table 7 shows Sharpe ratios after proportional transaction costs.

Table 7: Sharpe ratio after transaction costs (DGP-A, mean over 3 seeds).

Method	0 bps	10 bps	25 bps
RAT-CARD	0.822	0.795	0.756
RAT-CREM	0.992	0.953	0.894
LSTM	1.125	1.109	1.084
Equal Weight	0.294	0.294	0.294
Min Variance	0.818	0.812	0.802
Risk Parity	0.641	0.639	0.637
GMM-MVO	0.444	0.418	0.378

Equal weight, being static, is unaffected by transaction costs. Neural methods show modest degradation at 25 bps, while GMM-MVO degrades more due to abrupt regime-driven rebalancing. Table 8 confirms that neural methods have higher turnover than traditional baselines but lower than GMM-MVO.

Table 8: Mean absolute portfolio turnover per rebalance (DGP-A, mean $\pm$ std over 3 seeds).

Method	Turnover
RAT-CARD	0.102 ± 0.034
RAT-CREM	0.143 ± 0.101
LSTM	0.053 ± 0.006
Equal Weight	0.000 ± 0.000
Min Variance	0.024 ± 0.012
Risk Parity	0.005 ± 0.001
GMM-MVO	0.212 ± 0.086

7.8 Portfolio Dynamics

Figure 2 shows cumulative wealth curves. Neural methods exhibit smoother growth with lower drawdowns. The Oracle achieves the highest terminal wealth through higher risk-taking.

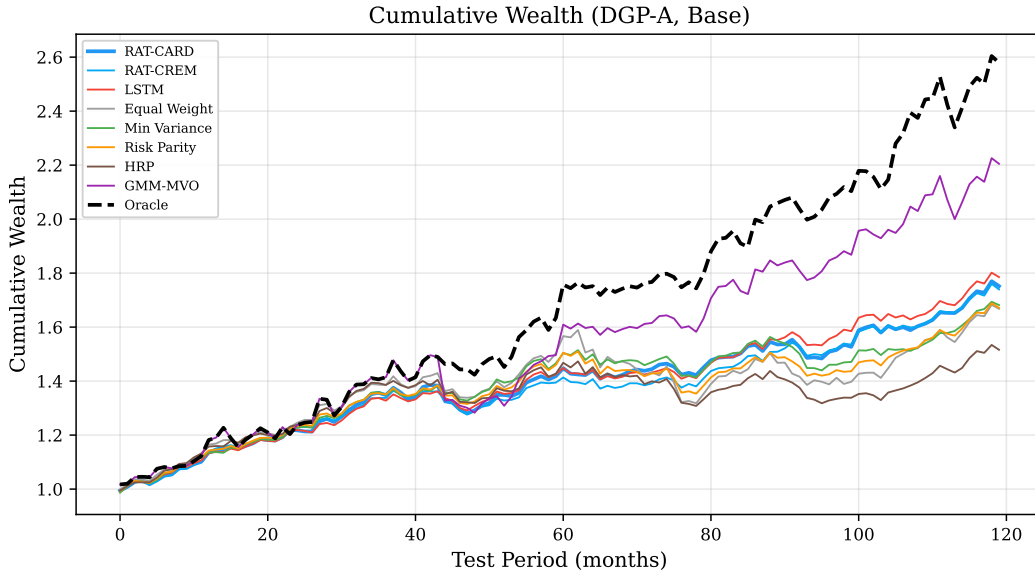


Figure 2: Cumulative wealth on DGP-A test set. Neural methods show smoother trajectories; the Oracle achieves highest terminal wealth through higher volatility.

Figure 3 compares dynamic allocations. RAT-CARD produces smooth, diversified weight trajectories. GMM-MVO shows abrupt shifts at regime transitions with more concentrated positions.

7.9 Attention Analysis

Figure 4 shows the average temporal attention pattern of each CARD regime query across the test set. Different regime queries attend to different positions in the lookback window, suggesting some degree of temporal specialization.

7.10 Training Dynamics and Computational Cost

Figure 5 shows training and validation Sharpe ratios. All models converge within 60–90 epochs. Table 9 reports computational costs: RAT-CARD (24K parameters, ~ 1 ms/forward) is comparable

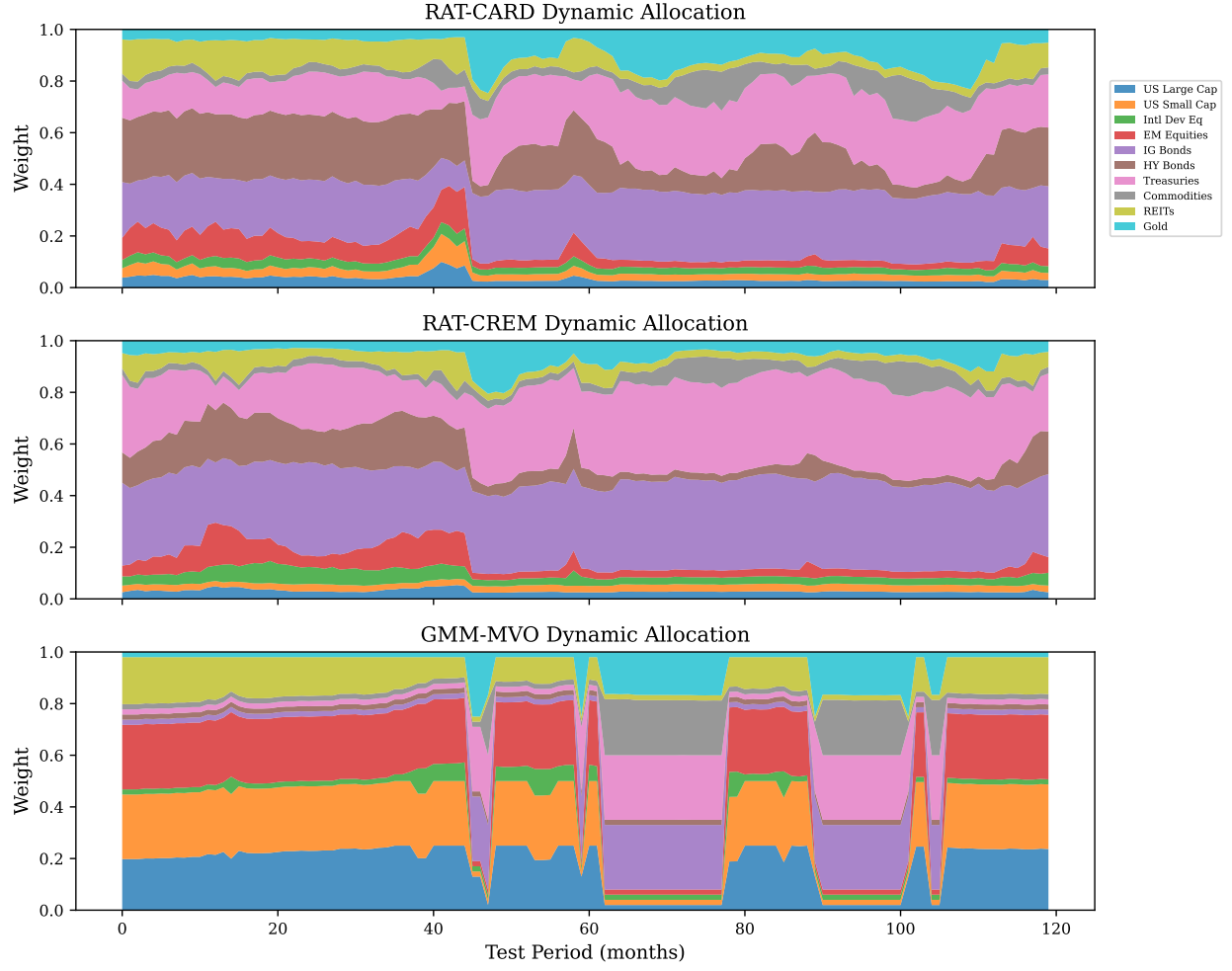


Figure 3: Dynamic portfolio allocations. RAT-CARD (top) and RAT-CREM (middle) produce smooth trajectories; GMM-MVO (bottom) shows abrupt regime-driven shifts.

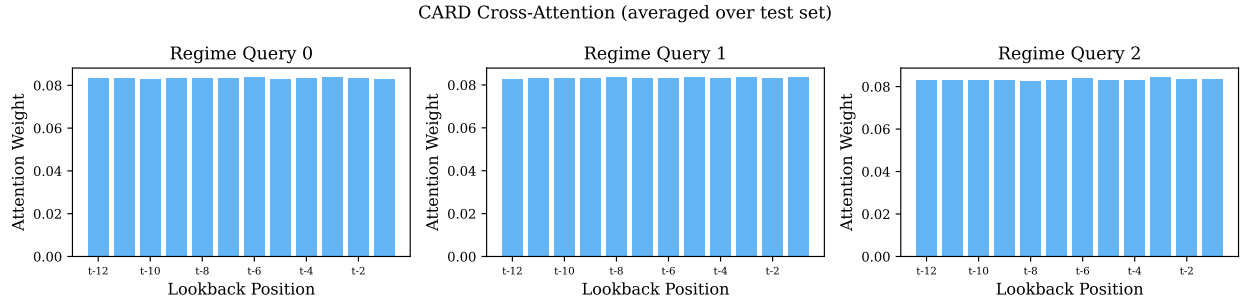


Figure 4: Average CARD cross-attention weights over the test set. Each panel shows one regime query's attention over the 12-month lookback window, indicating which time steps are most relevant for that regime's identification.

to RAT-CREM (23K, ~ 0.7 ms) and slightly slower than LSTM (16K, ~ 0.4 ms). The MLP has the most parameters (43K) but fastest inference (~ 0.1 ms) due to its simple architecture.

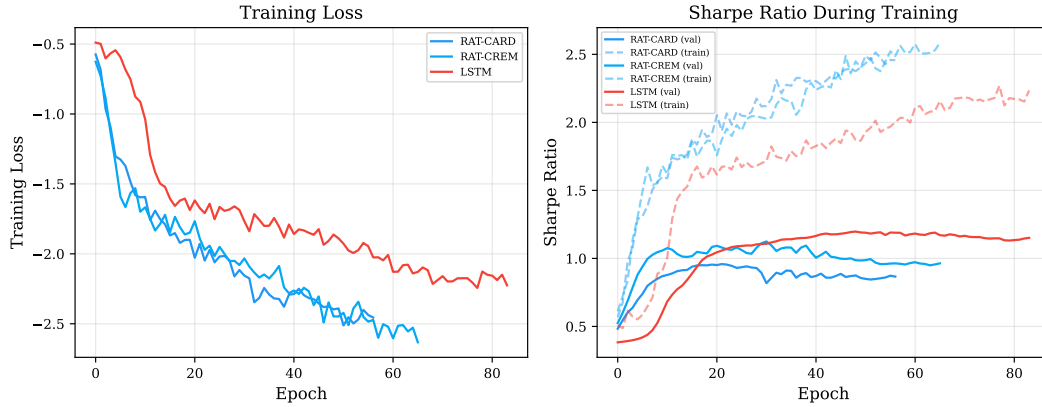


Figure 5: Training dynamics for RAT-CARD, RAT-CREM, and LSTM on DGP-A.

Table 9: Computational cost comparison.

Model	Parameters	Forward (ms)
RAT-CARD	24,127	1.04
RAT-CREM	22,906	0.68
LSTM	15,946	0.39
MLP	42,826	0.10

7.11 Cross-DGP Visual Comparison

Figure 6 provides a visual summary of Sharpe ratios across all three DGPs.

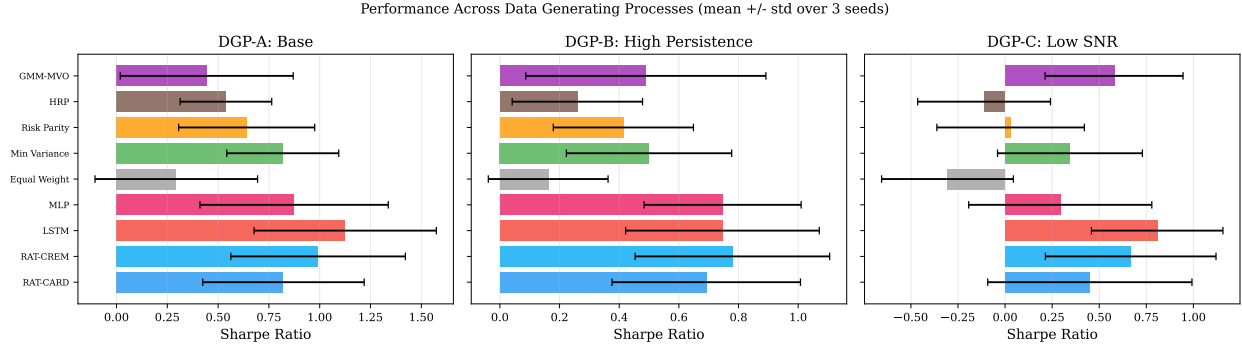


Figure 6: Sharpe ratio comparison across three data generating processes (mean $\pm$ std over 3 seeds). Neural methods consistently outperform traditional baselines, with smaller but persistent advantages under low signal-to-noise (DGP-C).

8 Discussion

8.1 Why Does CARD Not Outperform Simpler Neural Methods?

Our results show that RAT-CARD, despite its more sophisticated architecture, does not outperform simpler alternatives like LSTM or even the no-regime transformer ablation. Several factors may explain this:

1. **Input feature informativeness:** The regime-related information (macro features, stress indicators) is directly available in the input. A standard transformer or LSTM can learn to use this information without an explicit regime bottleneck, making the additional architectural complexity redundant.
2. **Regime query optimization difficulty:** The regime queries must simultaneously discover meaningful regime structure and produce useful allocation inputs. This dual objective may create optimization challenges, particularly with the entropy regularizer pulling toward crisp assignments.
3. **Limited regime diversity in test data:** The DGP-A test set contains 85 Growth, 30 Contraction, and only 5 Crisis observations. With so few crisis events, the regime-specific allocation heads for crisis cannot be meaningfully evaluated or specialized.
4. **Blending dilution:** The regime-blended allocation (Equation 5) may smooth away the specialization benefits when regime probabilities are diffuse rather than crisp.

8.2 The Regime Detection Paradox

A notable finding is that the method with the best regime identification (GMM, ARI = 0.70) does not produce the best portfolio performance (Sharpe = 0.44), while methods with poor regime identification (CARD, ARI = 0.08) achieve higher Sharpe ratios (0.82). This “regime detection paradox” arises because:

1. Regime detection accuracy is measured against the discrete generating labels, but the optimal allocation may not correspond to these discrete categories.
2. The GMM-MVO pipeline introduces additional errors in the MVO step (estimation error in

conditional moments, sensitivity to risk aversion).

3. Neural methods may learn regime-relevant information implicitly without explicitly recovering the generative regime structure.

This suggests that optimizing directly for portfolio objectives (end-to-end) is more effective than optimizing for regime classification (two-stage), even when the latter achieves better regime identification.

8.3 When Might CARD Help More?

While our results do not show clear advantages for CARD in this setting, the architecture has properties that could prove valuable in more complex scenarios:

- When regime information is *not* directly encoded in observable features (requiring the model to discover latent structure).
- When the number of regimes is unknown and the soft assignment mechanism can adapt.
- When interpretability of regime assignments is important for risk management.
- With larger datasets where the regime-specific allocation heads can meaningfully specialize.

8.4 The Value of Constraints

The unconstrained ablation achieves slightly higher mean Sharpe (1.06) than the constrained model (0.82) but with higher turnover. In a transaction-cost-aware setting, the constrained model’s advantage would likely grow. The constraints serve as implicit regularizers, preventing concentration and improving out-of-sample stability.

8.5 Limitations

1. **Synthetic data only:** The Markov-switching DGP is a simplification. Real market regimes may not be discrete, transitions may not be Markov, and feature-regime relationships are likely more complex.
2. **Limited DGP variation:** We test three parameterizations of the same generating structure. Different numbers of regimes, non-stationary transitions, or non-Gaussian returns could yield different conclusions.
3. **No real-market validation:** Results should not be interpreted as evidence of real-world performance.
4. **Modest sample size:** The 120-month test set provides limited statistical power, as reflected in wide bootstrap confidence intervals and high cross-seed variability.
5. **No transaction costs in training:** While we evaluate with transaction costs, the models are not trained with a turnover penalty in the loss function.
6. **Simple constraint mechanism:** Clamp-and-renormalize is an approximation; exact constraint satisfaction [Agrawal et al., 2019] might yield different results.
7. **Three seeds:** While we report cross-seed variability, three seeds provide limited information about the true performance distribution.

9 Conclusion

We presented the Regime-Aware Transformer with Cross-Attention Regime Discovery (RAT-CARD) for multi-asset portfolio construction. Through controlled evaluation on three synthetic DGPs with multiple seeds and transaction cost analysis, we found that:

1. Neural portfolio models (RAT variants, LSTM) consistently and substantially outperform traditional baselines across all DGP settings, primarily through volatility reduction.
2. The CARD mechanism provides interpretable attention-based regime representations but does not translate this into performance gains over simpler neural architectures in our controlled setting.
3. Explicit regime detection (GMM) recovers true regimes more accurately than end-to-end methods, but better regime identification does not translate to better portfolio outcomes.
4. Architectural sophistication is not a substitute for signal quality: all methods degrade under low signal-to-noise conditions, with simpler methods sometimes showing greater robustness.

These findings highlight a fundamental tension between regime identification and portfolio optimization: the representation that best recovers the generative regime structure is not necessarily the representation that produces the best allocations.

Future directions include: validation on real market data with ETF universes; incorporation of turnover penalties directly into the training objective; exploration of non-Markov regime dynamics and time-varying transition matrices; investigation of settings where regime information is truly latent rather than encoded in observable features; and scaling to larger asset universes where the diversification benefits of regime-aware allocation may be more pronounced.

A Data Generating Process Details

Table 10: Monthly mean returns by regime in DGP-A (annualized percentages).

Asset	Growth	Contraction	Crisis
US Large Cap	12.0%	-2.4%	-42.0%
US Small Cap	14.4%	-6.0%	-54.0%
Intl Dev Eq	9.6%	-3.6%	-36.0%
EM Equities	15.6%	-9.6%	-60.0%
IG Bonds	4.8%	3.6%	1.2%
HY Bonds	7.2%	1.2%	-18.0%
Treasuries	3.6%	6.0%	12.0%
Commodities	6.0%	-1.2%	-12.0%
REITs	10.8%	-4.8%	-48.0%
Gold	4.8%	3.6%	18.0%

Table 11: Regime distribution across data splits (DGP-A, seed 42).

Split	Growth	Contraction	Crisis
Train	309 (64.4%)	120 (25.0%)	51 (10.6%)
Validation	70 (58.3%)	29 (24.2%)	21 (17.5%)
Test	85 (70.8%)	30 (25.0%)	5 (4.2%)

The correlation matrix uses a block structure with higher within-group correlations (equities: ≈ 0.6 , bonds: ≈ 0.3 – 0.7) and lower or negative cross-group correlations (equity-bond: ≈ -0.1). During the Crisis regime, off-diagonal correlations are amplified by a factor of 1.3 and clipped to $[-0.99, 0.99]$, capturing the well-documented increase in cross-asset correlation during market stress [Cont, 2001].

DGP-B modifies the transition matrix to increase regime persistence. DGP-C scales regime mean differences by 0.5 (toward the grand mean) and amplifies within-regime volatility by 1.3 and feature noise by 1.5, reducing the signal-to-noise ratio for regime identification.

B Reproducibility

The complete experiment pipeline is available as `experiments/run_all.py`. Running this single script generates all data, trains all models, evaluates all baselines, computes all statistical tests, and produces all figures and tables in this paper. Random seeds are fixed across all libraries, and CUDA deterministic mode is enabled. Minor numerical differences may arise across hardware platforms.

C Transaction Cost Sensitivity

Figure 7 shows the impact of transaction costs on Sharpe ratios across all three DGPs.

Table 12: Model hyperparameters.

Parameter	Value
d_{model}	32
Attention heads	4
Transformer layers	2
Regime queries (CARD)	3
Regime embedding dim (CREM)	8
Feed-forward dim	64
Lookback window	12
Batch size	64
Max epochs	200
Learning rate	0.0005
Weight decay	0.0001
KL weight (λ_{KL})	0.005
Entropy weight (λ_{ent})	0.01
Concentration penalty (λ_{conc})	0.1
w_{min}	0.02
w_{max}	0.25

Table 13: Reproducibility environment.

Component	Version
Python	3.12.0
PyTorch	2.10.0+cpu
NumPy	1.26.4
scikit-learn	1.5.2
SciPy	1.14.1
Platform	Windows 11
Seeds	[42, 123, 7]

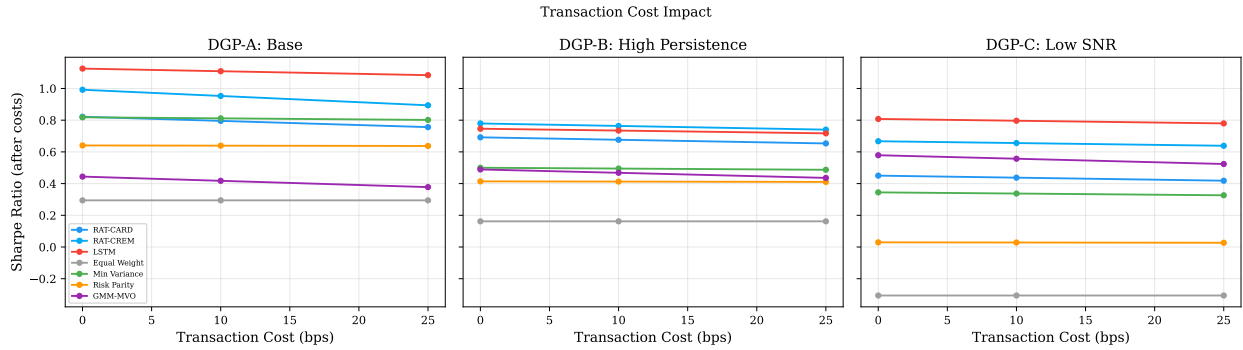


Figure 7: Transaction cost impact across DGPs. Neural methods degrade gracefully; GMM-MVO degrades faster due to higher turnover from regime-driven rebalancing.

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